Supplementary tables

Occupational injuries among foreign workers in Japan by residence status: industry- and age-standardized incidence ratios from public statistics, 2023–2025

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Tables S1–S5 are cited in the article. SIR, standardized incidence ratio; CI, confidence interval. The numbers are produced by analysis.py (see output/).

Table S1. Further sensitivity analyses: standardized incidence ratios (SIRs) adjusted for industry and age

Analysis	All foreign workers	Technical intern training	Specified skilled worker	Technical intern training + specified skilled worker	Status-based residents
Main analysis	1.25	1.27	1.94	1.43	1.66
Expected numbers summed over residence statuses	1.26	–	–	–	–
No employment weighting of age structure	1.21	1.27	1.94	1.43	1.53
Denominator: weight on previous year					
1/4	1.28	–	–	–	–
1/3	1.30	–	–	–	–
Age-adjustment factor[a] fixed at					
0.85	1.30	–	–	–	–
0.90	1.23	–	–	–	–
0.95	1.16	–	–	–	–
1.00 (no age adjustment)	1.10	–	–	–	–
Least favourable assumptions combined					
Agency workers in manufacturing and 10% more workers	1.08	–	–	–	–
The same plus age-adjustment factor 0.95	1.00	–	–	–	–
2023–2024 only					
Industry and age	–	1.26	1.90	1.39	1.66
Plus sex, same sex ratio in all industries	–	1.44	2.06	1.57	1.78
Plus sex, construction workers all men	–	1.22	1.89	1.35	1.71
Construction, if its workers are younger or older[b]					
Share aged 20–24 yr 8 points higher	–	0.92	1.79	1.03	–
8 points lower	–	1.00	1.99	1.12	–
12 points higher	–	0.90	1.74	1.01	–
12 points lower	–	1.02	2.05	1.15	–
Manufacturing, in the same four scenarios[b]					
Construction share 8 points higher	–	1.53	2.15	1.68	–
8 points lower	–	1.48	2.11	1.63	–
12 points higher	–	1.54	2.16	1.69	–
12 points lower	–	1.47	2.10	1.62	–
Medical, health care and welfare					
Compared with all workers in this industry	–	1.18	1.33	1.28	3.11
Compared with a mix of medical and care workers (74% care)	–	0.92	1.01	0.98	2.65
Compared with care workers only	–	0.76	0.83	0.81	2.26

–, not calculated. [a] Expected number adjusted for industry and age divided by the expected number adjusted for industry only (0.88–0.89 in the main analysis). [b] The main analysis assumes that workers of a residence status have the same age distribution in every industry. Here the share of construction workers aged 20–24 yr is raised or lowered by 8 or 12 percentage points (with the opposite change at 30–34 yr), and manufacturing is shifted the other way so that the overall age distribution is unchanged.

Table S2. Results for each year: number of injuries and standardized incidence ratio (SIR) adjusted for industry and age

Residence status	2023 Injuries SIR (95% CI)	2024 Injuries SIR (95% CI)	2025 Injuries SIR (95% CI)
All foreign workers	5,672 1.25 (1.22–1.28)	6,244 1.22 (1.19–1.25)	7,124 1.28 (1.25–1.31)
Specified skilled worker	597 1.98 (1.83–2.15)	810 1.84 (1.71–1.97)	1,177 2.00 (1.89–2.12)
Status-based residents	2,258 1.66 (1.60–1.73)	2,283 1.65 (1.58–1.72)	2,314 1.67 (1.60–1.74)
Other professional/technical	235 1.40 (1.23–1.60)	242 1.32 (1.16–1.50)	234 1.19 (1.05–1.36)
Technical intern training	1,692 1.30 (1.24–1.37)	1,874 1.23 (1.17–1.28)	2,046 1.30 (1.24–1.36)
Designated activities	161 1.03 (0.88–1.21)	171 0.93 (0.80–1.08)	258 1.13 (0.99–1.27)
Engineer/Humanities/International services	427 0.71 (0.65–0.78)	500 0.74 (0.68–0.81)	650 0.86 (0.80–0.93)
Permitted outside-status work	284 0.46 (0.41–0.52)	352 0.51 (0.46–0.57)	420 0.55 (0.50–0.61)
Technical intern training + specified skilled worker	2,289 1.43 (1.37–1.49)	2,684 1.36 (1.31–1.42)	3,223 1.49 (1.44–1.54)

In each cell, the first line is the number of injuries, the second the SIR and the third its 95% confidence interval (exact Poisson). All foreign workers: eight industries; includes injuries with unknown status.

Table S3. Observed and expected numbers of injuries by residence status, 2023–2025 pooled

Residence status	Observed	Expected, adjusted for industry	Expected, adjusted for industry and age	Observed minus expected
All foreign workers	19,040	17,246	15,248	3,792
Specified skilled worker	2,584	1,627	1,329	1,255
Status-based residents	6,855	4,430	4,129	2,726
Other professional/technical	711	688	547	164
Technical intern training	5,612	3,942	4,402	1,210
Designated activities	590	681	568	22
Engineer/Humanities/International services	1,577	2,849	2,029	−452
Permitted outside-status work	1,056	2,820	2,059	−1,003
Technical intern training + specified skilled worker	8,196	5,569	5,731	2,465

Observed minus expected uses the expected number adjusted for industry and age. All foreign workers: eight industries, including 55 injuries of unknown status; the rows for the residence statuses (six industries) therefore do not sum to this row.

Table S4. Standardized incidence ratios (SIRs) adjusted for age within each industry, 2023–2025 pooled

Industry	Technical intern training	Specified skilled worker	Technical intern training + specified skilled worker	Status-based residents	Engineer/Humanities/ International services
Manufacturing	1.50 (1.45–1.56)	2.13 (2.01–2.25)	1.66 (1.61–1.71)	2.95 (2.86–3.05)	1.48 (1.38–1.59)
Construction	0.96 (0.91–1.00)	1.88 (1.73–2.04)	1.07 (1.03–1.12)	1.96 (1.80–2.13)	0.66 (0.56–0.77)
Wholesale and retail trade	1.59 (1.40–1.78)	1.47 (1.19–1.79)	1.55 (1.40–1.72)	1.17 (1.07–1.27)	0.92 (0.81–1.03)
Medical, health care and welfare	1.18 (0.95–1.45)	1.33 (1.15–1.54)	1.28 (1.13–1.44)	3.11 (2.85–3.38)	0.84 (0.40–1.54)
Accommodation, eating and drinking services	0.87 (0.60–1.20)	1.09 (0.91–1.29)	1.03 (0.88–1.20)	0.77 (0.69–0.86)	0.74 (0.62–0.87)
All other industries	2.03 (1.87–2.19)	2.96 (2.67–3.27)	2.31 (2.17–2.45)	0.80 (0.76–0.85)	0.30 (0.26–0.34)

95% confidence intervals (exact Poisson) in parentheses. All other industries include agriculture and fishery, for which the number of foreign workers is not published by residence status; the estimates in this row for technical intern training and specified skilled worker cannot be interpreted (see Results).

Table S5. Age distribution (%) of all employees in Japan and of foreign workers by residence status used for standardization, 2025

Age (yr)	All employees in Japan	Specified skilled worker	Status-based residents	Other professional/technical	Technical intern training	Designated activities	Engineer/ Humanities/ International services	Permitted outside-status work
15–19	1.9	0.7	1.4	0.2	5.5	0.3	0.0	3.2
20–24	7.3	19.1	4.3	2.8	45.7	19.7	7.0	46.0
25–29	9.1	38.3	6.7	14.8	26.5	37.3	32.1	28.9
30–34	8.7	23.7	9.9	20.9	13.0	21.6	31.2	10.7
35–39	8.9	11.8	13.6	19.1	6.0	9.5	16.9	5.4
40–44	9.9	4.5	15.5	15.2	2.1	4.5	7.2	2.8
45–49	11.3	1.5	14.3	11.1	0.9	2.3	3.0	1.5
50–54	12.7	0.4	12.9	7.6	0.3	1.3	1.4	0.9
55–59	10.6	0.0	11.4	4.5	0.1	1.2	0.7	0.4
60–64	8.4	0.0	7.4	2.3	0.0	1.4	0.3	0.2
≥65	11.2	0.0	2.6	1.5	0.0	0.9	0.2	0.0

All employees in Japan: Labour Force Survey, 2025 average. Foreign workers: estimated from the age distribution of residents with each status at the end of June 2025; for status-based residents, designated activities and permitted outside-status work, residents were weighted by the ratio of employees to population in Japan (see Methods). The same distribution was applied to every industry. Columns may not sum to 100 because of rounding. Distributions for 2023 and 2024 are available in the public repository.